

12 GENERAL TABLE DATA

These cards are used only in NEW or RESTART type problems and are required only if any input references general tables. TTT is the table number, and table references such as for power, heat transfer coefficients, and temperatures refer to this number. Data must be entered for each table that is referenced, but TTT need not be consecutive. Tables entered but not referenced are stored, and this is not considered an error. During RESTART, general tables may be added, existing tables may be deleted, or existing tables may be modified by entering new data.

12.1 Table Type and Multiplier Data, 202TTT00

W1(A) Table type. In RESTART problems, DELETE can be entered to delete general table TTT. For non restart problems, the names in the following table can be used.

Table 12.1-1 Types of General Tables

Table Type	Quantity in Table
POWER	Power versus time
HTRNRATE	Heat flux versus time
HTC-T	Heat transfer coefficient versus time
HTC-TEMP	Heat transfer coefficient versus temperature
TEMP	Temperature versus time
REAC-T	Reactivity versus time or Control system variable versus time (prevents undesirable unit conversion)
NORMAREA	Normalized area versus normalized stem position

The following two, three, or four words are optional and allow trips and factors or units changes to be applied to the table entries. If the factors are omitted, the data are used as entered. One multiplier is used for time, power, heat flux, heat transfer coefficient, normalized stem position, and normalized area; a multiplier and additive constant are used for temperature as $T = M \cdot TX + C$, where M is the multiplier, C is the additive constant, and TX is the temperature entered. The first one or two factors apply to the argument variable, time or temperature; one factor is applied if the argument is time, and two factors are used if the argument is temperature. The remaining one or two factors are used for the function, two factors being used if temperature is the function. Note that the temperature can be either the argument (HTC-TEMP table type) or the function (TEMP table type). When factors are used for a table involving temperature as the argument, the first two factors apply to the temperature. The second and third factors apply to the temperature when temperature is the function.

W2(I) Table trip number. This number is optional unless factors are entered. If missing or 0, no trip is used, and the time argument in the following table is the time supplied to the table

for interpolation. If nonzero, the number is the trip number, and the time argument in the following table is -1.0 if the trip is false and the time supplied to the table minus the trip time if the trip is true. This field may be omitted if no factors are entered. This number must be 0 or blank for tables that are not a function of time.

W3-W5(R) Factors. As described above, enter factors such that when applied to the table values entered, the resultant values have the appropriate units. For the NORMAREA table, the resultant values for both the normalized length and area must be ≥ 0 and ≤ 1.0 .

12.2 General Table Data, 202TTT01-99

The card numbers need not be consecutive. The units given are the units required after the factors on card 202TTT00 have been applied. Pairs of numbers are entered; the limit on the number of pairs is 99.

W1(R) Argument value (s, if time; K, °F, if temperature; dimensionless, if normalized stem position).

W2(R) Function value (W, MW, if power; K, °F, if temperature; W/m^2 , $\text{Btu/s}\cdot\text{ft}^2$, if heat flux; $\text{W/m}^2\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}^2\cdot^\circ\text{F}$, if heat transfer coefficient; dollars, if reactivity; dimensionless, if normalized area).

The tables use linear interpolation for segments between table search argument values. For search arguments beyond the range of entered data, the end-point values are used.